SUPPLY CHAIN & PROFITABILITY ANALYSIS

Machine Learning Project Report

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This report presents a professional business analytics dashboard developed using Machine Learning concepts, Streamlit, Python, Pandas, and Plotly. The project focuses on customer behavior analysis, regional profitability, revenue trends, discount impact, and operational insights through interactive dashboard visualizations and exploratory data analysis.

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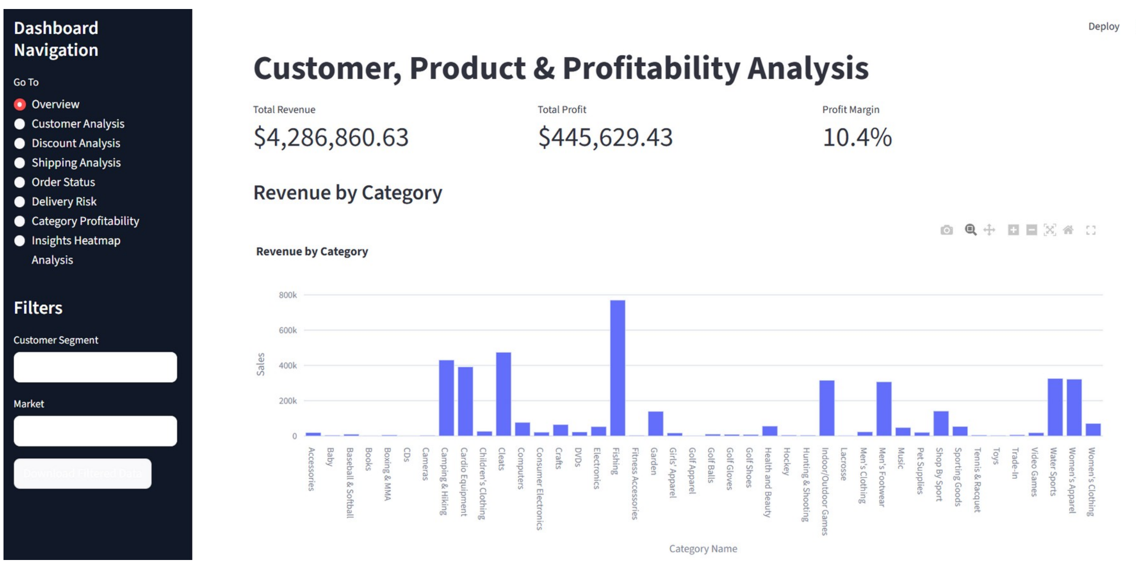
Executive Summary

This project focuses on building an interactive analytics dashboard for customer profitability, regional sales performance, product category analysis, discount impact evaluation, and operational insights using Machine Learning concepts and Business Intelligence techniques. The dashboard was developed using Streamlit, Python, Pandas, and Plotly to provide dynamic visual analytics and business-oriented insights.

The project demonstrates how data analytics and visualization can transform raw operational datasets into actionable insights that support strategic decision-making. The dashboard enables organizations to identify profitable customers, strong-performing regions, product-level opportunities, and operational risks related to shipping and delivery.

Figure 1: Interactive Supply Chain Analytics Dashboard



The analytical dashboard improves business intelligence capabilities by allowing users to interactively explore operational data, compare performance across markets, and identify profitability trends. The combination of data visualization and machine learning-oriented analysis supports strategic planning and operational optimization in modern business environments.

Introduction & Problem Statement

Modern businesses generate large volumes of sales, operational, and customer data every day. Without effective analytics systems, organizations struggle to identify profitable products, customer behavior patterns, operational inefficiencies, and growth opportunities.

This project addresses these challenges by creating a professional dashboard system capable of analyzing profitability trends, regional performance, customer contribution, shipping risks, and discount impact through interactive data visualizations.

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analysis supports strategic planning and operational optimization in modern business environments.

Dataset Overview

The dataset used in this project contains customer information, product categories, revenue values, profit metrics, shipping details, market regions, delivery status, and customer segment information. Data preprocessing and cleaning were performed using Pandas and NumPy libraries.

The analysis primarily focused on Pacific Asia market data and Consumer customer segments to generate meaningful profitability and operational insights.

Figure 2: Dashboard Architecture with Interactive Filters

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Revenue & Profit Analysis

Revenue and profit analysis identified strong-performing business areas and highlighted profitability variations across multiple operational dimensions. Revenue trend analysis revealed periodic spikes in business performance, while profit distribution analysis identified both profitable and loss-generating transactions.

The dashboard allows organizations to monitor business growth, identify profit drivers, and optimize operational strategies

Figure 3: Revenue Trend Analysis

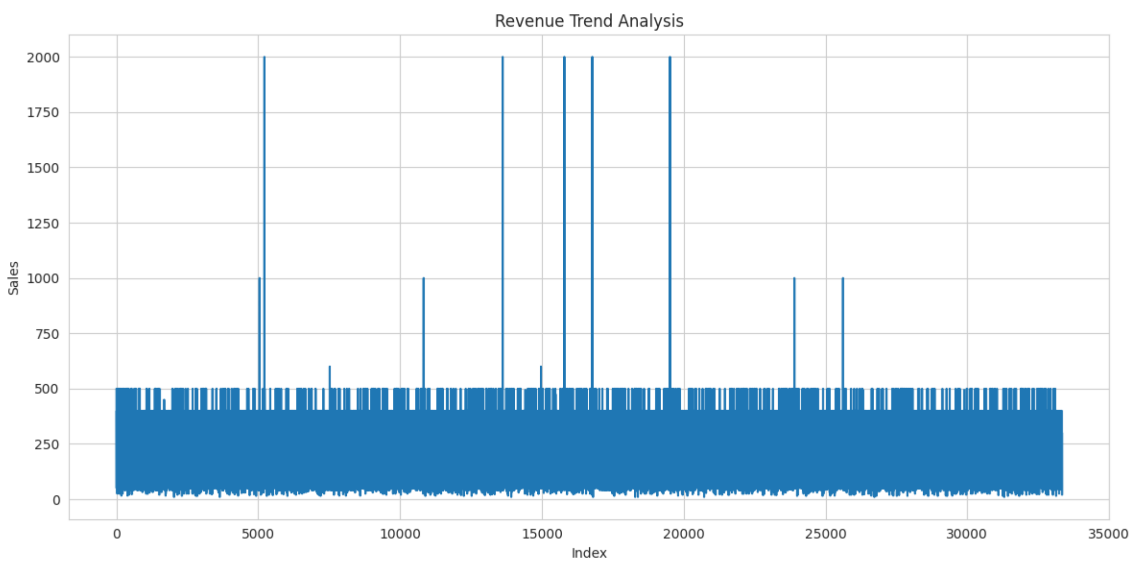


Figure 4: Profit Distribution Analysis

Customer Segment Analysis

Customer segmentation analysis showed that Consumer customers contributed the highest share of total revenue and business activity. Corporate customers maintained stable profitability while Home Office customers represented a smaller but valuable segment.
Top customer analysis identified specific customers generating exceptionally high profits, emphasizing the importance of customer retention strategies.

Figure 5: Customer Segment Contribution

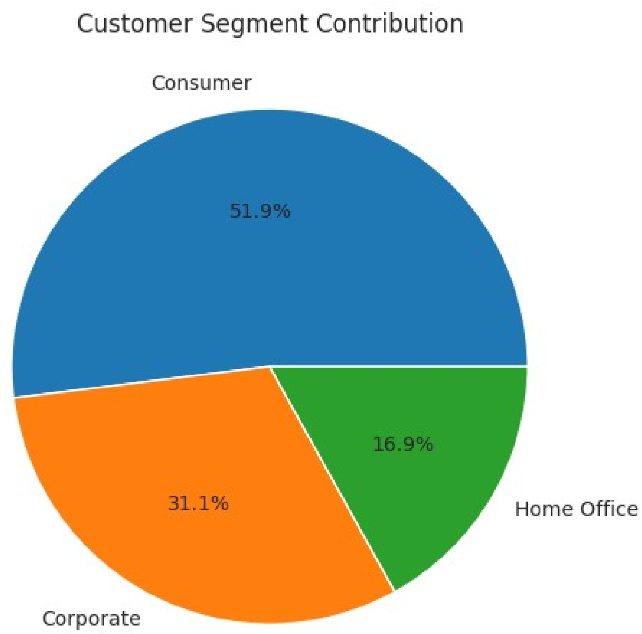
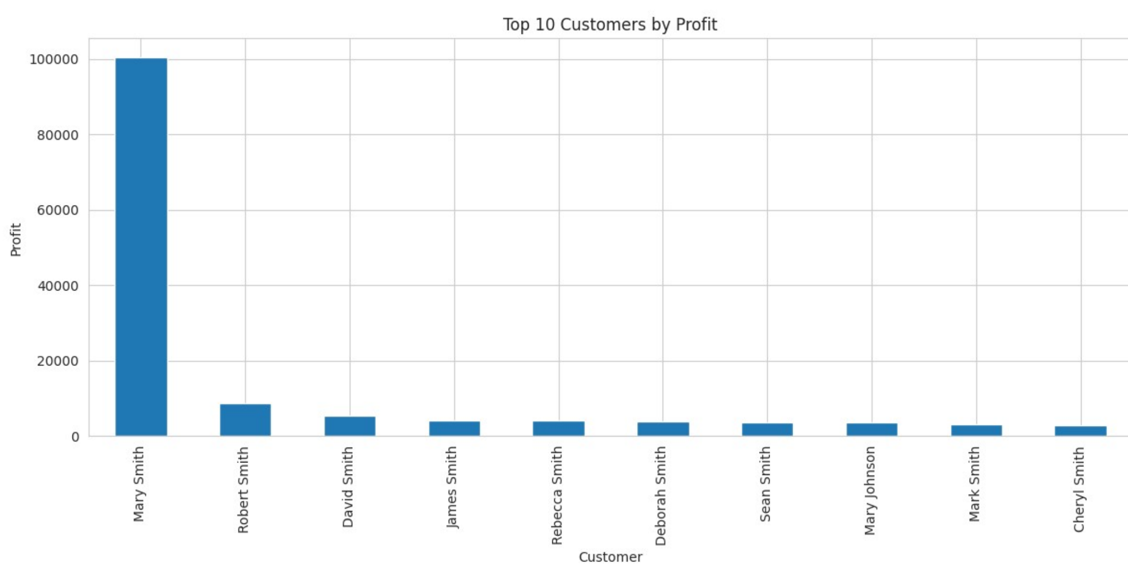


Figure 6: Top 10 Customers by Profit



Regional Profitability Analysis

Regional analysis demonstrated that Central America and Western Europe generated strong profitability, while Pacific Asia maintained stable operational performance. Some regions showed weaker profitability due to lower sales concentration and operational inefficiencies.

Regional comparisons support strategic expansion planning and operational optimization.

Figure 7: Regional Profitability Analysis

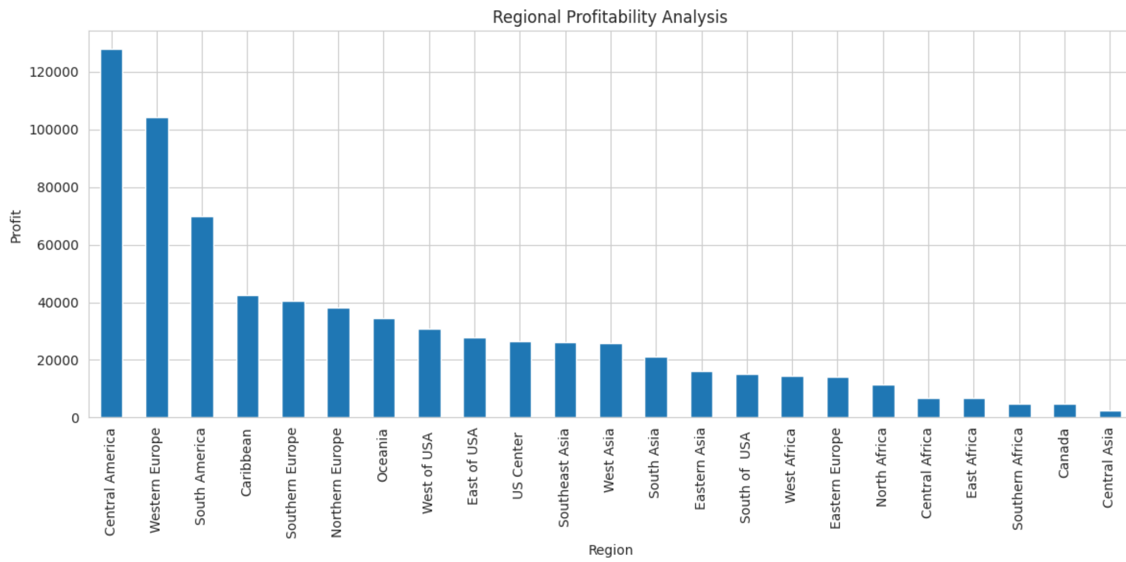
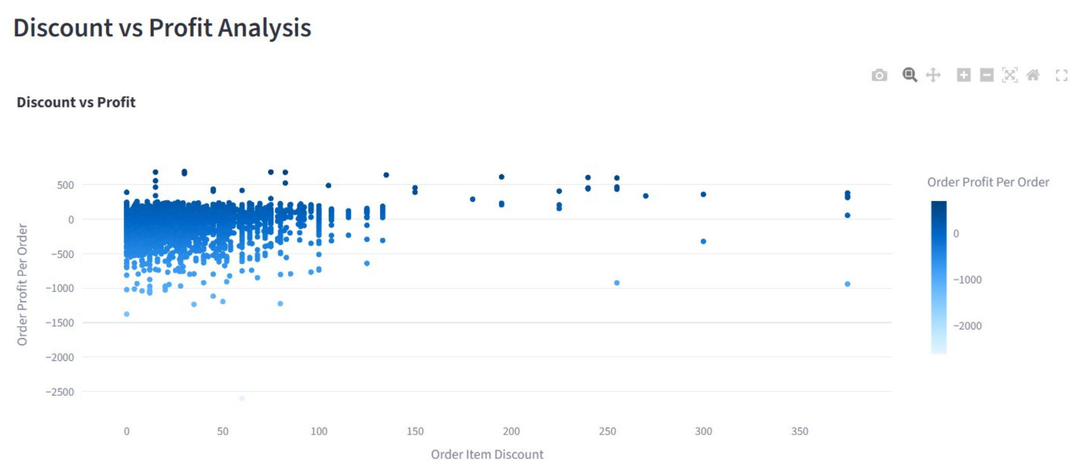


Figure 8: Pacific Asia Market Dashboard

Discount vs Profit Analysis

Discount analysis demonstrated a strong relationship between discount percentages and profitability. Excessive discounts frequently resulted in profit reduction and operational losses. Moderate discounting strategies appeared more sustainable for long-term growth and profitability maintenance.

Figure 9: Discount vs Profit Relationship



Product Category Performance

Product category analysis identified high-performing products and premium categories contributing significantly to total revenue and profitability. Certain categories generated strong sales volume while others delivered higher profit margins.

This analysis supports inventory optimization and product marketing strategies.

Figure 10: Product Profitability Analysis

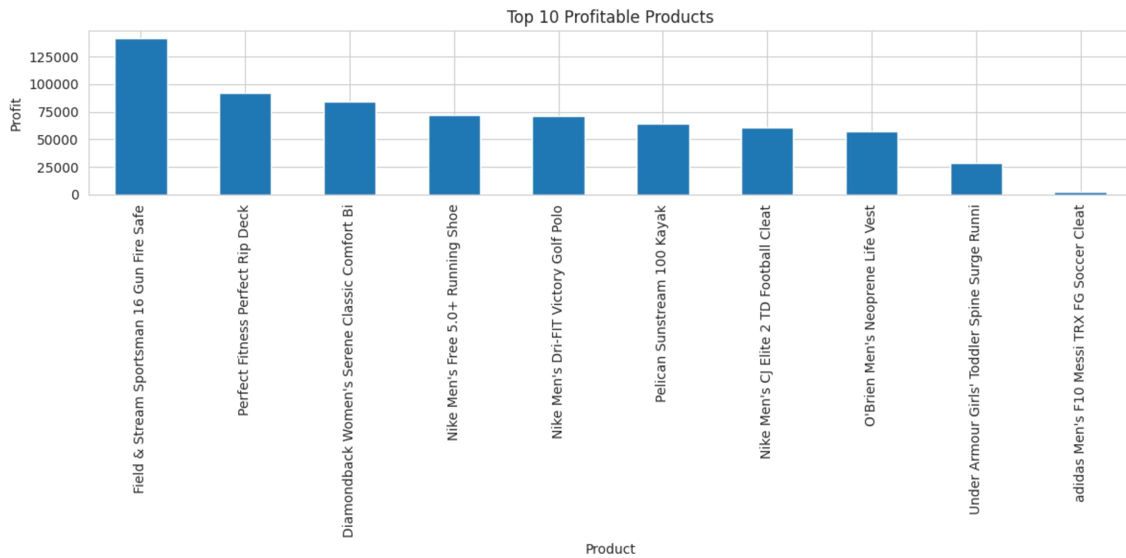
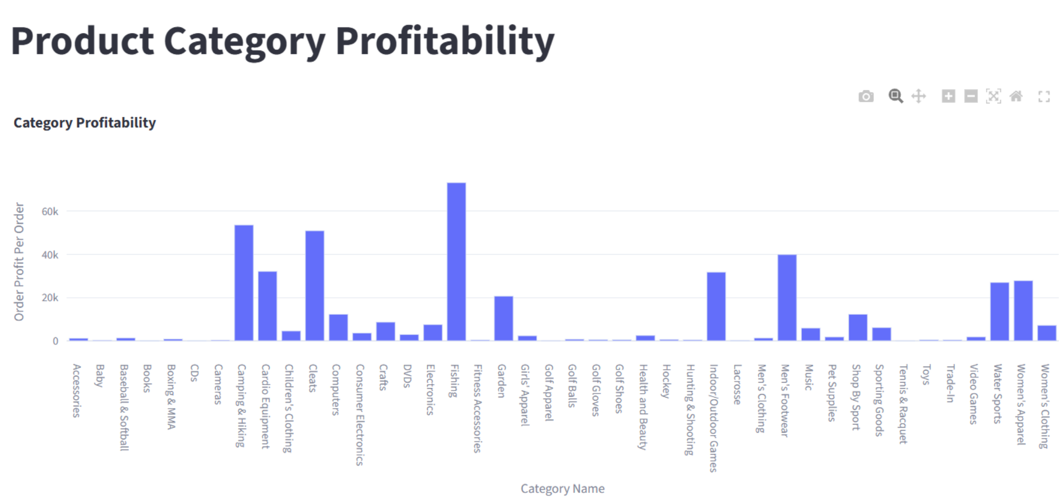


Figure 11: Product Category Performance



Shipping & Delivery Risk Analysis

Shipping and delivery analysis focused on operational efficiency and risk evaluation. The dashboard identified areas with higher delivery delays and shipping-related operational inefficiencies.

Shipping mode analysis helps organizations optimize logistics operations and improve customer satisfaction.

Figure 12: Shipping Mode Analysis

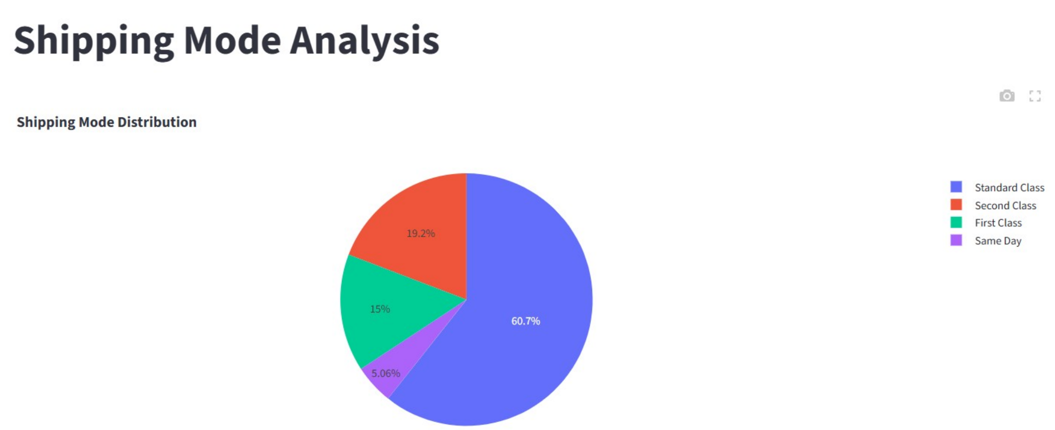
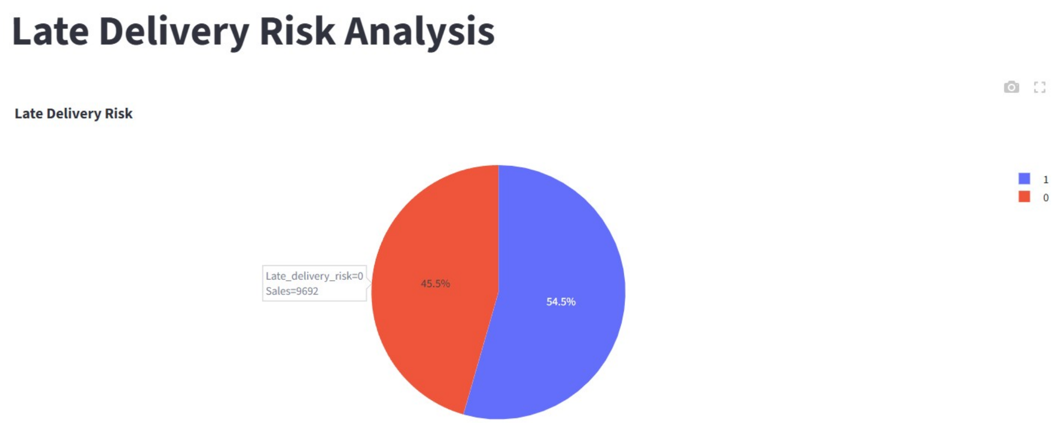


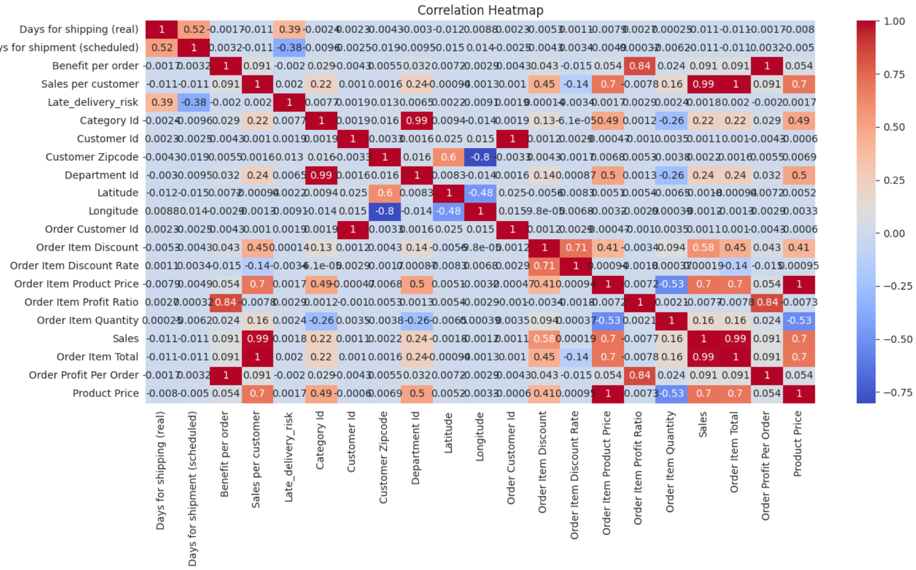
Figure 13: Late Delivery Risk Analysis



Heatmap & Correlation Analysis

Heatmap analysis was used to evaluate relationships between numerical variables within the dataset. The correlation matrix revealed dependencies between sales, profit, and operational variables, helping identify hidden business patterns and trends.

Figure 14: Feature Correlation Heatmap



Dashboard Features

The dashboard includes multiple advanced features such as interactive filters, KPI metrics, dark mode design, animated charts, CSV download functionality, and Plotly-based interactive visualizations.

These features improve usability and support dynamic analytical exploration.

Figure 15: Advanced Dashboard Features

Key Business Insights

1. Consumer customers contributed the highest overall revenue.
2. Excessive discounts negatively impacted profitability.

3. Certain regions generated disproportionately high profits.
4. A limited number of customers contributed major profit shares.
5. Shipping performance directly affected operational efficiency and customer satisfaction.
6. Product profitability varied significantly across categories.

Recommendations

- Optimize discount strategies to preserve profitability.
- Focus on high-value customer retention.
- Improve shipping efficiency in weaker-performing regions.
- Expand profitable product categories.

- Implement predictive analytics for future forecasting.
- Improve operational monitoring using business intelligence dashboards.

Future Scope

Future enhancements may include machine learning forecasting models, recommendation systems, inventory optimization, real-time database integration, and AI-based business intelligence systems.

The dashboard can evolve into a complete predictive analytics platform for enterprise-level operational management.

Conclusion

Project success Summary ✔

This project successfully demonstrates the practical application of Machine Learning concepts, Data Analytics techniques, and Business Intelligence methodologies for solving real-world business problems.

The dashboard provides meaningful insights regarding customer profitability, revenue trends, regional performance, discount impact, product analysis, and shipping risks. Interactive visualizations and analytical dashboards improve decision-making capabilities and support profitability optimization through data-driven strategies.

Overall, the project highlights the importance of analytics systems in transforming complex business datasets into understandable and actionable insights.

References

Python for Data Analysis – Wes McKinney

Hands-On Machine Learning – Aurélien Géron

Plotly Documentation

Streamlit Documentation

Pandas Official Documentation

Business Intelligence & Data Analytics Concepts

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